

SHIVANSH SINGH

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EDUCATION

University of Toronto | BAsC - Electrical & Computer Engineering

Graduated April 2026

- **Minors:** Artificial Intelligence Engineering | Robotics & Mechatronics
- **Relevant Coursework:** Probability & Applications (ECE302), Applied Linear Algebra (MAT188), Algorithms & Data Structures (ECE345), Operating Systems (ECE344), Computer Security (ECE568), Machine Learning (ECE421, CSC384, APS360), Linear Control Systems (ECE410), Software Design & Communication (ECE297), Fundamentals of Accounting & Finance (JRE300)

TECHNICAL SKILLS

Programming: Modern C++ (17/20/23), Python, TypeScript/JavaScript (React, Node.js), Go, SQL (PostgreSQL), Java, C (bare-metal), x86/ARM assembly

Systems & Architecture: Distributed Systems, Lock-Free Concurrency, Real-Time / Embedded, API Design (REST, gRPC), Zero-Copy Data Paths

Robotics, Controls & ML: ROS2, PX4, Simulink, PID / closed-loop control, Inverse Kinematics, Visual Servoing, OpenCV, HITL/SITL Simulation, PyTorch, CUDA, OpenVINO, YOLOv5, GANs

Hardware: STM32, Arduino, Raspberry Pi, PCB Design, FPGA, UART/BLE, Servo & Motor Control, 3D Printing

Tools & Infrastructure: Docker, Git, CI/CD, AWS (EC2, S3), Linux, Valgrind, GDB, Intel VTune

WORK EXPERIENCE

Software Engineering Intern, Robotics | Ribbit (Autonomous Cargo Aircraft — NRC search-and-rescue mission)

Sep. 2024 – Aug. 2025

- Engineered real-time C++17 service layers using zero-copy data pathing and lock-free ring buffers, sustaining sub-100ms deterministic message-processing cycles.
- Wrote embedded drivers for safety-critical sensors/actuators with template metaprogramming for zero-overhead hardware abstraction.

Robotics Intern | Roboprenr

May 2023 – Aug. 2023

- Built a voice-interactive pseudo-humanoid robot from zero: 3D-printed chassis + custom PCBs wiring sensors, servos (PCA9685), and IMU to a Raspberry Pi 4.
- ROS2 Humble on embedded Linux for modular control; OpenCV vision in C++ and PyTorch face-recognition models.

TEAMS & LEADERSHIP

Software Lead - UAV Multicopter | UofT Aerospace Team (UTAT)

Sep. 2025 – Apr. 2026

- Architected a resilient, event-driven IPC framework in Modern C++ for autonomous flight nodes — fault-tolerant crash resilience and asynchronous state synchronization across mission-critical cycles.
- PX4 flight control for multi-copter/fixed-wing VTOL mode transitions; closed-loop attitude + rate controllers in Simulink with PID gain tuning from flight-log analysis.

Software Developer - Deep Learning Acceleration | aUToronto (UofT Self-Driving Team)

Sep. 2023 – Apr. 2024

- Refactored latency-critical perception pipelines from Python to Modern C++ ROS2 nodes, eliminating GIL constraints — 2×+ throughput.
- Profiled with Intel VTune and Valgrind to fix cache misses and heap bottlenecks — ~5× faster inference via memory-efficient ops, ~25% lower CPU at peak load.

PROJECTS

Agentfail: Empirical Failure-Mode Study of Agent Sessions | Python (stdlib only), Wilson Score Intervals

Aug. 2026

- Measured how coding agents actually fail across **85,104 tool calls in 1,854 real agent runs** (228 main sessions + 1,626 subagents, 30.2 days): 3.34% [3.22–3.46] tool-call error rate, with agents adapting after an error in 96.9% of cases rather than retrying blindly (1.76%).
- Built a replication harness re-running every analysis against a **different agent framework** — **94,059 tool calls across 13 models** — and reported which findings survived; replication overturned one of my own headline claims and the writeup says so.

Placebo: Inert-Code Static Analysis | Python, Typer, AST Detectors

Aug. 2026

- Finds *inert code* — present, reviewed, merged, believed to work, and provably without effect. Evaluated on **104 held-out repositories (110,550 files, 28.8M lines)** at **30.0% precision**, after an honest recalibration from a tuned-set number that did not generalize (100% tuned → 16.5% unseen → 30.0% fixed).

- Found a real defect in a shipped library: SQLAlchemy 2.1.0b4's `IdentitySet.__ixor__` computes the symmetric difference, discards it, and returns `self` — confirmed by executing it.
- Indexes 130k LOC in 4.4 s and the CPython 3.10 standard library (460k LOC) in 18.4 s.

Tracon: Dependency-Aware Scheduler for Agentic Workloads | C++ *kernel*, Go/gRPC via *cgo*, Python *DES Simulator* Jul. 2026 – Present

- One C++ scheduling kernel compiled twice — a pybind11 module the simulator calls in-process and a Go gRPC service — measured at **1–52 μ s per decision in-process, 215–417 μ s over gRPC** by queue depth.
- Trace-driven simulator validated against recorded traces at **0.0% median / 0.1% p90 replay error** over 2,489 turns, and **0.0000% median and maximum error across 13 external runs** (3,894 agent runs, 101,766 model steps, 243.6 h of model service).

Agent Radar [Marketplace] | TypeScript, VS Code Extension API Jul. 2026 – Present

- Published to the Visual Studio Marketplace: surfaces every AI coding agent running on the machine — Claude Code sessions and subagents, Codex, Copilot CLI, Cursor Agent, Grok — in the status bar and a live dashboard webview.
- **Fully local by design** — no server, no network; agent discovery scans processes and session transcripts rather than any vendor API. Per-CLI source adapters behind one collector interface, with a completion-detection precedence chain resolving conflicting signals.

Daisy: Agent Runtime over a Git-Versioned Vault | Claude Agent SDK, MCP, Cloudflare Workers, TypeScript, Python Jun. 2026 – Present

- File-based personal operating system — every note is Markdown + YAML under git, so every agent mutation is an attributable, revertible commit and the audit trail is the version history.
- Multi-agent orchestration layer of ~ 27 composable skills, including a seat system mirroring a software team (supervisor $\rightarrow$ dev $\rightarrow$ parallel worker subagents $\rightarrow$ review panel) with git-worktree isolation so agents build concurrently without collisions.

Lockwire: A Physics Lockfile for Robots | Python, Typer, pydantic, NumPy, PX4 ULog May 2026 – Present

- CLI that pins a robot's **physical** assumptions — masses, power rails, bus bandwidth, actuator torque — into a checked-in contract and fails CI when they stop adding up, plus a go/no-go preflight gate the powered-on robot must pass before it operates.
- Every check compares **two independently authored sources** (datasheet vs. datasheet, URDF joint limit vs. actuator torque $\times$ gear ratio, Σ message rate $\times$ payload vs. bus bitrate) — verifying records against each other rather than one author's arithmetic against their own guesses.

3rdspace: Presence-Based Campus Social App | SwiftUI, Core Location, Supabase (Postgres/RLS) Feb. 2026 – Present

- iOS app built end-to-end: geofenced hotspots, anonymous-until-friended identity, private and public spaces across 195 Swift source files.
- Designed the privacy model as an engineering constraint rather than a policy page — Postgres row-level security as the enforcement layer, plus a sealed safety log with break-glass-only access and a 12-month rolling delete.

Autonomous Robotic Service Dog (Capstone, Lead Architect) [Repo] [Demos] | ROS2, Nav2, C++, OpenCV Sep. 2025 – Apr. 2026

- Led the system architecture for an 8-DOF quadruped that follows and monitors patients — distributed ROS2 stack integrating perception, planning, and control with custom QoS profiles for deterministic multi-node communication.
- Built a closed-loop visual-servoing pipeline (OpenCV state estimates $\rightarrow$ velocity commands) enabling adaptive real-time human tracking; end-to-end person-following demonstrated on the physical quadruped.

Rescue Ranger [Repo] [Devpost] | C++, OpenCV, Node.js, AWS (EC2, S3), React, Arduino Jan. 2023

- Built a first-place autonomous search-and-rescue rover (1st of 150+ teams, UofTHacks X) in 24 hours: 3D-printed chassis, Arduino, C++ obstacle avoidance, and OpenCV survivor identification.
- Node.js REST backend on AWS EC2 streaming live rover telemetry (media in S3), powering a real-time React + Google Maps dashboard.

Additional Projects 2022 – 2026

- **KUKA KR6 Arm** (2025): 6-DOF — DH kinematics, FK/IK, jerk-limited trajectories (Simulink). **STM32 BLE Rover** (2024): bare-metal C — obstacle avoidance + BLE camera streaming (< 500 ms), UART, FPGA, assembly.
- **TourGuide Pro** (ECE297, 2023): multithreaded C++ OpenStreetMap engine — Dijkstra/A*, ~ 200 ms queries. **Security Exploits** (ECE568, 2024): buffer overflows; XSS, SSRF, RCE, prototype pollution on CTF-style targets.
- **Orphan** (Aug. 2026): macOS scanner for data still held by installed apps that no longer reference it — Trash-only deletion, protected paths, no **--force**, 144 tests. **Vyuha** (Jul. 2026): deterministic co-op battle simulator in TypeScript — seeded RNG and fixed-tick determinism enforced by test.
- **PacBot** (UTRA, 2023): autonomous Pac-Man AI — grid search + strategy. **Project Jerome** (MakeUofT 2022) [Devpost]: wearable obstacle + gas-hazard detection for the visually impaired. **Accent Ace** (HackTheNorth 2023) [Devpost]: PyTorch/torchaudio pronunciation coach.

EXTRACURRICULARS

Hackathons: UofTHacks X 1st Place, UTEK 2nd Place, HackTheNorth 2023 [Devpost], NSBEHacks [Devpost]

Varsity Athlete: UofT Wrestling & Rowing; MMA/BJJ bronze medallist (Canadian nationals); former national-level swimmer (India). 15+ hrs/week competitive training alongside full-time engineering.